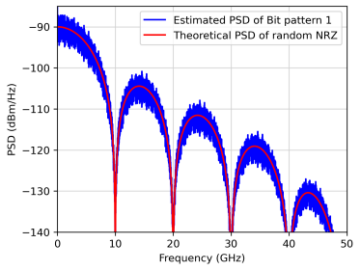
Homework: Spectrum of digital signals

Three bit-pattern files are provided to you, each contains 200,000 bits. Estimate the PSDs of the three cases using the averaged periodogram method. Details:

- The bit rate is 10 Gbps; risetime = 20 ps (trapezoidal wave); V_{pp} = 1 V; Z_0 = 50 Ω. You need to first construct the time-domain waveform based on these parameters (and the bit pattern from the files), before you can apply the periodogram method. This part is the same as in HW3.
- The resolution of the estimated PSD should be at least (smaller than) 10 MHz. This sets a minimum truncation time constraint in your calculation. The plotting Δf should be $\leq$ 1 MHz.
- Add the power at negative frequencies to the positive frequencies, so that the results are comparable to spectrum analyzer measurements.
- Plot the estimated PSD against the theoretical PSD of purely random NRZ on the same figure.
- X-axis: 0~50 GHz. Y-axis: in the unit of dBm/Hz.
- Use two (arbitrary) different values of K for each case. (Total 6 PSD plots)
- You can use any window functions. Remember to compensate the window gain. You are encouraged to try using coherent and incoherent window gains to see the effects.

Example answer:

	Bit pattern 1	Bit pattern 2	Bit pattern 3
K = 10			
K =			

In fact, one of the three bit-patterns is an 8b/10b-coded random signal, one is a purely random NRZ, and one is a special hybrid-random-periodic pattern. In addition to the six PSD plots, answer the following questions:

1. Determine which file is which pattern.
2. Look at the description of 8b/10b in Wikipedia and then give an explanation for the PSD of 8b/10b you obtain. What property of 8b/10b gives rise to such spectrum?
3. For the special hybrid-random-periodic pattern, based on the PSD and the pattern file, explain the reason for the PSD you obtain.

Notes:

1. As discussed in the class note, the theoretical PSD of purely random NRZ is

$$S_{NRZ}(f) = 2 \cdot \frac{V_{\text{pp}}^2 T_b}{4Z_0} \text{sinc}^2(fT_b) \, \text{sinc}^2(fT_r) \qquad \text{W/Hz}$$

   The leading factor 2 accounts for the power at negative frequencies.

2. For the purely random bit pattern, your estimated PSD should agree with the theoretical PSD very well when K ≥ 100. This serves as a check for the correctness of your code.

Provide your Python source code.